# MATHEMATICS AND STATISTICS

**Time: 3 Hrs.** **Max. Marks: 80**

***General instructions:***

*The question paper is divided into **FOUR** sections.*

*(1)* ***Section A:*** *Q. 1 contains **Eight** multiple choice type questions, each carrying **Two** marks. Q. 2 contains **Four** very short answer type questions, each carrying **One** mark.*

*(2)* ***Section B:*** *Q.3 to Q. 14 contain **Twelve** short answer type questions, each carrying **Two** marks. (Attempt any **Eight**)*

*(3)* ***Section C:*** *Q.15 to Q.26 contains **Twelve** short answer type questions, each carrying **Three** marks. (Attempt any **Eight**)*

*(4)* ***Section D:*** *Q.27 to Q.34 contains **Eight** long answer type questions, each carrying **Four** marks. (Attempt any **Five**)*

*(5) Use of log table is allowed. Use of calculator is not allowed.*

*(6) Figures to the right indicate full marks.*

*(7) Use of graph paper is not necessary. Only rough sketch of graph is expected.*

*(8) For each multiple choice type of questions, only the first attempt will be considered for evaluation.*

*(9) Start answer to each section on a new page.*

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## SECTION – A

**Q.1. Select and write the correct answer for the following multiple choice type of questions:** **[16]**

i. If A = {1, 2, 3, 4, 5} then which of the following is not true? (2)

(A) $\exists\, x \in A$ such that $x + 3 = 8$ (B) $\exists\, x \in A$ such that $x + 2 < 9$

(C) $\forall\, x \in A$, $x + 6 \geq 9$ (D) $\forall\, x \in A$, $x + 5 \leq 10$

ii. The value of $\sin\left[\sin^{-1}\left(-\frac{1}{\sqrt{2}}\right) - 3\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right] =$ (2)

(A) $-\frac{1}{\sqrt{2}}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $-\frac{\sqrt{3}}{2}$

iii. If $\hat{p}, \hat{q}, \hat{r}$ are mutually perpendicular unit vectors and form a right handed triplet then the value of $\hat{p}\cdot(\hat{q} \times \hat{r}) + \hat{q}\cdot(\hat{p} \times \hat{r}) + \hat{r}\cdot(\hat{p} \times \hat{q}) =$..... (2)

(A) –2 (B) 0 (C) 1 (D) 3

iv. The Cartesian equation of the line passing through the points A(3, 4, –1) and B(2, –1, 3) is _____. (2)

(A) $\frac{x+3}{1} = \frac{y-4}{5} = \frac{z+1}{-4}$ (B) $\frac{x-3}{-1} = \frac{y-4}{-5} = \frac{z+1}{-4}$

(C) $\frac{x-3}{-1} = \frac{y-4}{-5} = \frac{z+1}{4}$ (D) $\frac{x-3}{-1} = \frac{y-4}{5} = \frac{z+1}{-4}$

v. If $y = \log_x a$ then find $\frac{dy}{dx} =$ _____. (2)

(A) $\frac{-\log a}{x(\log x)^2}$ (B) $x \log a$ (C) $-\frac{1}{\log_x a}$ (D) $\frac{1}{x \log a}$

vi. If $f(x) = \frac{\sin^{-1}x}{\sqrt{1-x^2}}$ and $g(x) = e^{\sin^{-1}x}$ then

$\int f(x)\cdot g(x)\, dx =$ _____. (2)

(A) $e^{\sin^{-1}x}(1 - \sin^{-1}x) + c$ (B) $e^{\sin^{-1}x}(\sin^{-1}x - 1) + c$

(C) $e^{\sin^{-1}x}(\sin^{-1}x + 1) + c$ (D) $e^{\cos^{-1}x}(\cos^{-1}x - 1) + c$

vii. The area bounded by the line $y = 2x$, X – axis and the lines $x = -1$, $x = 4$ is _____sq. units (2)

(A) 15 (B) 17 (C) 16 (D) 14

viii. In a Binomial Distribution, E(X) = 6, Var(X) = 4.2 then the value of n is _____. (2)

(A) 22 (B) 24 (C) 20 (D) 26

**Q.2. Answer the following questions:** **[4]**

i. Write the dual of $p \wedge t$. (1)

ii. What is the distance of point (3, –5, 6) from XZ plane? (1)

iii. The displacement of a particle at time t is given by $S = 3t^2 + 4t + 1$. Find its velocity at time t. (1)

iv. Write the order of differential equation $\left(\frac{d^2y}{dx^2}\right)^{\frac{5}{2}} = \sqrt[3]{\frac{dy}{dx}}$ (1)

## SECTION – B

**Attempt any EIGHT of the following questions:** **[16]**

**Q.3.** Construct the switching circuit for the statement $(\sim p \wedge q) \vee (p \wedge \sim r)$. (2)

**Q.4.** Find the matrix of cofactors of matrix $\begin{bmatrix} 5 & 2 \\ -1 & 6 \end{bmatrix}$. (2)

**Q.5.** Find the general solution of equation $\cos 5\theta = \frac{1}{2}$ (2)

**Q.6.** Find the value of k, if $2x + y = 0$ is one of the lines represented by $3x^2 + kxy + 2y^2 = 0$. (2)

**Q.7.** If $\bar{a}, \bar{b}, \bar{c}$ are the position vectors of points A, B, C respectively and $10\bar{a} = 7\bar{b} + 3\bar{c}$ then find the ratio in which the point C divides the line segment AB. (2)

**Q.8.** Find the angle between the lines
$\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(2\hat{i} - \hat{j} + \hat{k})$
$\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \mu(\hat{i} + 2\hat{j} + \hat{k})$ (2)

**Q.9.** Test whether the function $f(x) = x - \frac{1}{x}$, $x \in R$, $x \neq 0$ is increasing or decreasing. (2)

**Q.10.** Evaluate: $\int_{-1}^{1} |x| \, dx$ (2)

**Q.11.** Evaluate: $\int x \log x \, dx$ (2)

**Q.12.** Find the area bounded by curve $x^2 + y = 0$, X – axis and lines $x = 1, x = 4$. (2)

**Q.13.** The probability distribution of r.v. X is as follows:

| X = x | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| P(X = x) | 0.1 | k | 2k | 2k | k |

Find: (i) k (ii) F(2) (2)

**Q.14.** Evaluate: $\int \frac{e^x(1+x)}{\cos^2(xe^x)} \, dx$ (2)

## SECTION – C

**Attempt any EIGHT of the following questions:** **[24]**

**Q.15.** Using truth table, determine whether statement pattern $[(p \vee q) \wedge \sim p] \wedge \sim q$ is a tautology or a contradiction or contingency. (3)

**Q.16.** Prove that:
$\sin^{-1}\left(\frac{3}{5}\right) + \cos^{-1}\left(\frac{12}{13}\right) = \sin^{-1}\left(\frac{56}{65}\right)$ (3)

**Q.17.** In $\Delta ABC$, if a = 10, b = 8, c = 6 then find the value of
(i) $\sin\left(\frac{C}{2}\right)$ (ii) $\cos\left(\frac{B}{2}\right)$ (3)

**Q.18.** Are the four points A(1, 2, 1), B(2, –3, 4), C(3, 4, –5), D(2, 3, –2) coplanar? Justify your answer. (3)

**Q.19.** Find the distance between parallel lines.

$\frac{x}{2} = \frac{y}{-1} = \frac{z}{2}$ and

$\frac{x+1}{2} = \frac{y-1}{-1} = \frac{z+1}{2}$ (3)

**Q.20.** Find the vector equation of the plane passing through the points A(2, 1, 1), B(0, 2, 3), C(4, 5, 6). (3)

**Q.21.** If $y = \log\left[\frac{x+\sqrt{x^2+a^2}}{\sqrt{x^2+a^2}-x}\right]$, find $\frac{dy}{dx}$. (3)

**Q.22.** A wire of length 36 cm is bent to form a rectangle. Find its dimensions, if the area of the rectangle is maximum. (3)

**Q.23.** Verify Rolle's theorem for the function $f(x) = x^2 - 5x + 9$, $x \in [1, 4]$ (3)

**Q.24.** Solve the differential equation $x\frac{dy}{dx} - \sqrt{x^2+y^2} = y$ (3)

**Q.25.** The following is the p.d.f. of continuous r.v.

$f(x) = \frac{x}{8}$; $0 < x < 4$

$= 0$, Otherwise

Find (i) expression for c.d.f of X.

(ii) F(x) at 0.5 and 5 (3)

**Q.26.** If the p.m.f of r.v. X be

$P(X = x) = {}^4C_x \left(\frac{5}{9}\right)^x \left(\frac{4}{9}\right)^{4-x}$ for $x = 0, 1, 2, 3, 4$

$= 0$ ; otherwise

then find E(X) and Var (X). (3)

## SECTION – D

**Attempt any FIVE of the following questions:** **[20]**

**Q.27.** Find the inverse of matrix $\begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix}$ (4)

**Q.28.** Prove that the homogeneous equation of degree two in $x$ and $y$, $ax^2 + 2hxy + by^2 = 0$ represents a pair of lines passing through the origin if $h^2 - ab \geq 0$. (4)

**Q.29.** Prove that if $\bar{a}, \bar{b}, \bar{c}$ are three non coplanar vectors, then any vector $\bar{r}$ in the space can be uniquely expressed as a linear combination of $\bar{a}, \bar{b}, \bar{c}$. (4)

**Q.30.** Solve the linear programming problem (L.P.P.) by graphical method.

Minimize: $z = 8x + 10y$

Subject to, $2x + y \geq 7$

$2x + 3y \geq 15$

$x \geq 0$,

$y \geq 2$ (4)

**Q.31.** If $x = f(t)$ and $y = g(t)$ are differentiable functions of t, so that $y$ is differentiable function of $x$ and $\frac{dx}{dt} \neq 0$ then prove that $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$.

Hence find $\frac{dy}{dx}$, if $x = \sin t$, $y = \cos t$. (4)

**Q.32.** Evaluate: $\int \frac{2}{3 + 2\sin^2 x + 5\cos^2 x}\, dx$ (4)

**Q.33.** Prove that,

$\int_{-a}^{a} f(x)\, dx = 2\int_0^a f(x)\, dx$, if $f(x)$ is even

$= 0$, if $f(x)$ is odd. (4)

**Q.34.** A body cools according to Newton's law from 100°C to 60°C in 20 minutes. The temperature of the surrounding being 20°C. How long will it take to cool to 40°C? (4)